

PREMIUM RESOURCE · AI & FINANCE AUTOMATION

AI Output Validation Framework

A simple framework for validating AI-generated financial outputs before use — covering number checks, logic checks, and source verification.

3-Part Framework

Number Check Protocol

Logic Check Guide

Premium · €29

3

check types

6

number checks

5

logic checks

5

source checks

HOW THIS PAIRS WITH THE HALLUCINATION RISK GUIDE

This category's Hallucination Risk Guide explains why AI output can be confidently wrong and what that looks like in finance-specific scenarios. This framework is the practical companion — the actual step-by-step process to run before relying on a specific piece of AI-generated financial content, whatever risk produced the error in the first place.

SECTION 1

Validation Framework Overview

Three distinct check types, each catching a different kind of error — and a simple way to decide how much validation a given output actually needs.

The Three Check Types

Check Type	What It Catches	Most Critical When
Number Checks	Arithmetic errors, wrong scale or units, figures pulled from the wrong period.	The output contains any specific, quotable figure.
Logic Checks	Plausible-sounding but unsupported explanations, internal inconsistency, correlation mistaken for causation.	The output explains why something happened, not just what happened.
Source Verification	Fabricated or misattributed citations, quotes taken out of context, outdated references.	The output cites a policy, contract, regulation, or external document.

Deciding How Much Validation an Output Needs

Not every AI output deserves the same depth of scrutiny. Match the effort to the actual stakes — a quick internal draft doesn't need the same treatment as something reaching a board or an external party.

L

Light Touch

Internal, low-stakes, easily reversible

A quick internal draft, a first-pass summary you'll personally review before anyone else sees it, or an exploratory question with no immediate decision attached. A reasonableness check on any headline number is usually enough.

S

Standard

Internal but decision-influencing

Content that will inform a real internal decision, even if it stays inside the team — a variance explanation feeding a resourcing call, a forecast used for internal planning. Apply the full Number Check Protocol and a basic logic check.

F

Full Validation

External, board-facing, or high-stakes

Anything reaching a board, auditor, regulator, investor, or external party, or feeding a decision with real financial or reputational consequences. Apply all three check types in full, with a named human sign-off before use.

Common pitfall: Applying Light Touch validation by default and only escalating after something has already gone wrong. Decide the tier before you start working with the output, based on where it's headed — not after you've already gotten comfortable with a confident-sounding draft.

SECTION 2

Number Check Protocol

Six steps to run on any specific, quotable figure in AI-generated financial content before it's used for anything that matters.

1 Locate the number in the actual source

Find the exact figure in the original data, document, or system — not a similar-looking number, and not from memory of what it “usually” is.

EXAMPLE: *An AI states Q3 revenue was \$4.2M — open the actual GL export and confirm \$4.2M is really there, not \$4.02M or \$4.2M for a different quarter.*

2 Confirm scale and units match

Check whether the figure reflects the right unit — thousands versus millions, percentage versus absolute value — since this is a common silent error.

EXAMPLE: *A variance stated as “12%” needs checking against whether the source data was already in percentage terms or whether that's a derived figure.*

3 Recompute any derived figure independently

Never trust AI-performed arithmetic. Recalculate any sum, variance percentage, ratio, or growth rate yourself, or through a real formula.

EXAMPLE: *If the AI states a variance of “+8.3%,” calculate (actual–budget)/budget yourself from the source figures rather than accepting the stated percentage.*

4 Cross-check against an independent control total

Confirm the number reconciles with something that should independently agree with it — do the parts sum to the stated whole?

EXAMPLE: *If five departmental figures are claimed to sum to a stated company total, add them yourself and confirm they actually do.*

5 Confirm the time period matches what was asked

Check the number is genuinely for the period requested, not an adjacent period the model defaulted to.

EXAMPLE: *A request for “this month's” figure returning last month's actual, because the current month isn't yet closed in the source system.*

6 Apply a basic reasonableness check

Ask whether the number passes a simple gut check against history, budget, or a known benchmark before treating it as correct.

EXAMPLE: *A stated expense figure that's 10x the same line item's typical monthly value warrants a second look before anything else.*

SECTION 3

Logic Check Guide

Five ways to validate the reasoning behind an AI-generated explanation, not just the numbers it cites.

1 Trace the causal chain step by step

Confirm each step in the AI's explanation actually follows from the one before it, rather than jumping to a plausible-sounding conclusion.

EXAMPLE: *"Revenue increased because retention improved" skips the actual step of showing retention data — ask what specifically in the data supports that link.*

2 Check internal consistency across the output

Confirm different parts of the same response agree with each other — a stated total should match the sum of components mentioned elsewhere in the same answer.

EXAMPLE: *If the summary states a total of \$500K but the individual line items mentioned later add to \$480K, that inconsistency needs resolving before either figure is used.*

3 Test for an equally plausible counter-explanation

Ask whether an alternative explanation would fit the same data just as well — if so, treat the AI's explanation as one hypothesis, not a confirmed finding.

EXAMPLE: *A margin decline attributed to "input cost inflation" might equally be explained by a pricing change — check which one the actual data supports.*

4 Separate correlation from causation

Check whether the AI is claiming that one thing caused another when the data only shows the two moved together in the same period.

EXAMPLE: *Marketing spend and revenue both rose in the same quarter — that alone doesn't establish the spend caused the revenue increase.*

5 Check the premise, not only the conclusion

If the AI's starting facts are wrong, the reasoning built on them doesn't matter how internally sound it appears — verify the premise first.

EXAMPLE: *An explanation built on "given that Q2 was a low base" needs that base-period claim checked before evaluating anything that follows from it.*

A useful test: Read the explanation with every number removed — does the remaining logical structure still make sense on its own? If the reasoning only sounds convincing because of the specific figures attached, the underlying logic may be weaker than it first appears.

SECTION 4

Source Verification Steps

Five steps to confirm any citation, quote, or external reference is real, current, and accurately represented.

- 1 Locate the exact source document or system**

Identify specifically where the cited claim is supposed to come from, by name, not a general sense of “somewhere in the policy.”

EXAMPLE: *“Per the expense policy” needs a specific section number located in the actual current policy document, not just accepted as stated.*
- 2 Confirm the specific claim is actually present**

Check that the source genuinely contains the claim as stated — not just something similar or plausible given the topic.

EXAMPLE: *If a summary claims a contract includes a specific clause, open the contract and confirm that clause exists in that specific wording.*
- 3 Check that context was preserved**

A technically accurate quote can still misrepresent the source if the surrounding context changes its actual meaning.

EXAMPLE: *A quoted sentence that sounds like an approval might actually be describing a rejected option earlier in the same document.*
- 4 Verify the source is current, not outdated**

Confirm a cited policy, rate, or regulation is the version currently in effect, not a superseded version the model happened to reference.

EXAMPLE: *A cited expense limit needs checking against the policy's current effective date, not just its existence somewhere in a document.*
- 5 Confirm the source is actually authoritative**

Check that the cited source is genuinely the right or official reference for this claim, not an informal or secondary source being treated as definitive.

EXAMPLE: *A finance figure attributed to “the system” should specify which system — a report built from the ERP and a shadow spreadsheet can disagree.*

Common pitfall: Verifying that a citation exists without verifying that it says what was claimed. A real section number attached to a wrong paraphrase is just as risky as a citation that doesn't exist at all — both steps matter.

PUTTING IT TOGETHER

Quick Reference Checklist

One consolidated pass across all three check types — for Standard or Full Validation tier content, before it goes anywhere.

- ☐ Every specific figure has been located in its actual source, not just recognised as plausible.
- ☐ Any derived number (a sum, percentage, or ratio) has been independently recalculated, not accepted as stated.
- ☐ The stated explanation's causal chain has been traced step by step, not just accepted because it sounds reasonable.
- ☐ Any part of the output that should agree with another part actually does.
- ☐ Every citation or quote has been checked against its actual source, including whether context was preserved.
- ☐ A named human has signed off on the content before it reaches its intended audience, appropriate to its validation tier.

THE ONE-SENTENCE VERSION

Every specific, checkable claim in AI-generated financial content — a number, an explanation, or a citation — gets checked against its actual source before use, with the depth of checking matched to what's actually at stake.

Premium resource from MDD Business Consulting. Licensed for individual professional use. Not for redistribution or resale.

BUILDING A VALIDATION HABIT ACROSS YOUR TEAM

For a facilitated working session embedding this framework into your team's actual review process and tooling — a bespoke engagement is available on request.

info@mddbusinessconsulting.com